BagmatiCyber Ltd.

Linux Role-Based Access Control (RBAC) Lab

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Date :- April 2026

Overview

This document provides a complete step-by-step guide for configuring Role-Based Access Control (RBAC) on a Linux system for BagmatiCyber Ltd. It covers user/group management, directory permissions, sudo policies, cron jobs, and cleanup procedures.

Department	Group Name	Users
Development	dev_team	Jatil, Lok (primary)
Audit & Compliance	audit_team	Mahesh, Lok (secondary)
Operations	ops_team	Suchana

Part 1: User and Group Setup

1.1 Create Groups

Create the three department groups:

```
┌──(kali㉿kali)-[~]
└─$ sudo groupadd dev_team

┌──(kali㉿kali)-[~]
└─$ sudo groupadd audit_team

┌──(kali㉿kali)-[~]
└─$ sudo groupadd ops_team
```

1.2 Create users with correct primary and secondary groups

Also create personal groups (same name as user)

Create user groups

```
┌──(kali㉿kali)-[~]
└─$ sudo groupadd Jatil

┌──(kali㉿kali)-[~]
└─$ sudo groupadd Mahesh

┌──(kali㉿kali)-[~]
└─$ sudo groupadd Lok

┌──(kali㉿kali)-[~]
└─$ sudo groupadd Suchana
```

Create users with primary group

```
┌──(kali㉿kali)-[~]
└─$ sudo useradd -g dev_team -G Jatil Jatil

┌──(kali㉿kali)-[~]
└─$ sudo useradd -g audit_team -G Mahesh Mahesh

┌──(kali㉿kali)-[~]
└─$ sudo useradd -g dev_team -G audit_team,Lok Lok

┌──(kali㉿kali)-[~]
└─$ sudo useradd -g ops_team -G Suchana Suchana
```

Set passwords:

```
┌──(kali㉿kali)-[~]
└─$ sudo passwd Jatil

New password:
Retype new password:
passwd: password updated successfully

┌──(kali㉿kali)-[~]
└─$ sudo passwd Mahesh

New password:
Retype new password:
passwd: password updated successfully

┌──(kali㉿kali)-[~]
└─$ sudo passwd Lok

New password:
Retype new password:
passwd: password updated successfully

┌──(kali㉿kali)-[~]
└─$ sudo passwd Suchana
New password:
Retype new password:
passwd: password updated successfully
```

1.3 Verify Group Membership

```
┌──(kali㉿kali)-[~]
└─$ id Jatil

uid=1001(Jatil) gid=1001(dev_team) groups=1001(dev_team),1004(Jatil)

┌──(kali㉿kali)-[~]
└─$ id Mahesh

uid=1002(Mahesh) gid=1002(audit_team) groups=1002(audit_team),1005(Mahesh)

┌──(kali㉿kali)-[~]
└─$ id Lok

uid=1003(Lok) gid=1001(dev_team) groups=1001(dev_team),1002(audit_team),1006(Lok)

┌──(kali㉿kali)-[~]
└─$ id Suchana
uid=1004(Suchana) gid=1003(ops_team) groups=1003(ops_team),1007(Suchana)
```

Part 2: Directory and File Setup

2.1 Create Directory Structure

```
┌──(kali㉿kali)-[~]
└─$ sudo mkdir -p /opt/Bagmaticyber/projects

┌──(kali㉿kali)-[~]
└─$ sudo mkdir -p /opt/Bagmaticyber/logs
```

2.2 Create Required Files

```
┌──(kali㉿kali)-[~]
└─$ echo "Developer tasks for Q3" | sudo tee /opt/Bagmaticyber/projects/dev_notes.txt

Developer tasks for Q3

┌──(kali㉿kali)-[~]
└─$ echo "Audit trail begins here" | sudo tee /opt/Bagmaticyber/logs/audit.log
Audit trail begins here

┌──(kali㉿kali)-[~]
└─$ echo "Bagmati Access Control Policy" | sudo tee /opt/Bagmaticyber/README.md
Bagmati Access Control Policy
```

Part 3: Ownership and Permission Configuration

3.1 Set File Ownership

```
┌──(kali㉿kali)-[~]
└─$ sudo chown Jatil:dev_team /opt/Bagmaticyber/projects/dev_notes.txt

┌──(kali㉿kali)-[~]
└─$ sudo chown Mahesh:audit_team /opt/Bagmaticyber/logs/audit.log

┌──(kali㉿kali)-[~]
└─$ sudo chown root:ops_team /opt/Bagmaticyber/README.md
```

3.2 Set Directory Permissions + SetGID

The /projects directory must be accessible only to its owner and dev_team members. The /logs directory must be accessible only to its owner and audit_team members. Both use SetGID for group inheritance.

```
┌──(kali㉿kali)-[~]
└─$ sudo chown root:dev_team /opt/Bagmaticyber/projects

┌──(kali㉿kali)-[~]
└─$ sudo chmod 2770 /opt/Bagmaticyber/projects

┌──(kali㉿kali)-[~]
└─$ sudo chown root:audit_team /opt/Bagmaticyber/logs

┌──(kali㉿kali)-[~]
└─$ sudo chmod 2770 /opt/Bagmaticyber/logs
```

Path	Owner	Group	Permissions	Special Bit
/projects/dev_notes.txt	Jatil	dev_team	640	—
/logs/audit.log	Mahesh	audit_team	640	—
/README.md	root	ops_team	444	—
/projects/ (dir)	root	dev_team	770	SetGID (2)
/logs/ (dir)	root	audit_team	770	SetGID (2)

3.3 README.md permissions

```
┌──(kali㉿kali)-[~]
└─$ sudo chmod 444 /opt/Bagmaticyber/README.md
```

Part 4: sudo Configuration

4.1 Add Suchana to sudo Group

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG sudo Suchana
```

4.2 Configure Fine-Grained sudo Policies

Create a dedicated sudoers drop-in file for Suchana (safer than editing /etc/sudoers directly):

```
┌──(kali㉿kali)-[~]
└─$ sudo visudo
```

Enter the following content:

```
Suchana ALL=(ALL) NOPASSWD: /bin/systemctl restart networking
Suchana ALL=(ALL) !/bin/su, !/bin/bash
```

4.3 Validate sudo Configuration

```
┌──(kali㉿kali)-[~]
└─$ su Suchana
Password:
$ sudo systemctl restart networking
$ sudo su
[sudo] password for Suchana:
Sorry, user Suchana is not allowed to execute '/usr/bin/su' as root on kali.
$ sudo bash
[sudo] password for Suchana:
Sorry, user Suchana is not allowed to execute '/usr/bin/bash' as root on kali.
$
```

Part 5: Testing & Validation

Verification Commands

```
┌──(kali㉿kali)-[~]
└─$ su Lok
Password:
$ echo "test" >> /opt/Bagmaticyber/logs/audit.log
sh: 1: cannot create /opt/Bagmaticyber/logs/audit.log: Permission denied
$

┌──(kali㉿kali)-[~]
└─$ su Mahesh
Password:
$ ls /opt/Bagmaticyber/projects
ls: cannot open directory '/opt/Bagmaticyber/projects': Permission denied
$

┌──(kali㉿kali)-[~]
└─$ su Suchana
Password:
$ sudo systemctl restart networking
$

┌──(kali㉿kali)-[~]
└─$ su Suchana
Password:
$ sudo su
[sudo] password for Suchana:
Sorry, user Suchana is not allowed to execute '/usr/bin/su' as root on kali.
$

┌──(kali㉿kali)-[~]
└─$ su Jatil
Password:
$ rm /opt/Bagmaticyber/README.md
rm: remove write-protected regular file '/opt/Bagmaticyber/README.md'?
$

┌──(kali㉿kali)-[~]
└─$ su Lok
Password:
$ touch /opt/Bagmaticyber/logs/test.txt
$
```

Part 6: Group Reassignment & Cleanup

6.1 Reassign Lok to dev_team Only

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -G dev_team Lok
```

6.2 Remove Suchana from sudo Group

```
┌──(kali㉿kali)-[~]
└─$ sudo deluser Suchana sudo
```

6.3 Remove All Secondary Groups — Reset to Personal Group Only

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -g users -G "" Jatil

┌──(kali㉿kali)-[~]
└─$ sudo usermod -g users -G "" Mahesh

┌──(kali㉿kali)-[~]
└─$ sudo usermod -g users -G "" LOk
usermod: user 'LOk' does not exist

┌──(kali㉿kali)-[~]
└─$ sudo usermod -g users -G "" Lok

┌──(kali㉿kali)-[~]
└─$ sudo usermod -g users -G "" Suchana
```

6.4 Delete All Users (Including Home Directories)

```
┌──(kali㉿kali)-[~]
└─$ sudo userdel -r Jatil
userdel: group Jatil not removed because it is not the primary group of user Jatil.
userdel: Jatil mail spool (/var/mail/Jatil) not found
userdel: Jatil home directory (/home/Jatil) not found

┌──(kali㉿kali)-[~]
└─$ sudo userdel -r Mahesh
userdel: group Mahesh not removed because it is not the primary group of user Mahesh.
userdel: Mahesh mail spool (/var/mail/Mahesh) not found
userdel: Mahesh home directory (/home/Mahesh) not found

┌──(kali㉿kali)-[~]
└─$ sudo userdel -r Lok
userdel: group Lok not removed because it is not the primary group of user Lok.
userdel: Lok mail spool (/var/mail/Lok) not found
userdel: Lok home directory (/home/Lok) not found

┌──(kali㉿kali)-[~]
└─$ sudo userdel -r Suchana
userdel: group Suchana not removed because it is not the primary group of user Suchana.
userdel: Suchana mail spool (/var/mail/Suchana) not found
userdel: Suchana home directory (/home/Suchana) not found
```

6.5 Delete Groups and Directories

```
┌──(kali㉿kali)-[~]
└─$ sudo groupdel dev_team


┌──(kali㉿kali)-[~]
└─$ sudo groupdel audit_team

┌──(kali㉿kali)-[~]
└─$ sudo groupdel ops_team

┌──(kali㉿kali)-[~]
└─$ sudo groupdel Jatil

┌──(kali㉿kali)-[~]
└─$ sudo groupdel Mahesh

┌──(kali㉿kali)-[~]
└─$ sudo groupdel Lok

┌──(kali㉿kali)-[~]
└─$ sudo groupdel Suchana
```

```
┌──(kali㉿kali)-[~]
└─$ sudo rm -rf /opt/Bagmaticyber
```

Part 7: Cron Jobs and Aliases

7.1 Scheduled Log Backup for Mahesh

Configure an hourly cron job that creates a timestamped backup of audit.log:

```
┌──(kali㉿kali)-[~]
└─$ sudo crontab -u Mahesh -e
```

Add the following line:

```
0 * * * * cp /opt/Bagmaticyber/logs/audit.log /opt/Bagmaticyber/logs/archive/audit_$(date +\%F_\%H-\%M).log
```

Verify cron was saved

```
┌──(kali㉿kali)-[~]
└─$ sudo crontab -u Mahesh -l
```

Ensure archive directory exists and Mahesh can write to it:

```
┌──(kali㉿kali)-[~]
└─$ sudo chown Mahesh:audit_team /opt/Bagmaticyber/logs/archive

┌──(kali㉿kali)-[~]
└─$ sudo chmod 770 /opt/Bagmaticyber/logs/archive
```

7.2 Custom Alias for Lok

Create a persistent alias devread that lets Lok quickly read dev_notes.txt:

```
┌──(kali㉿kali)-[~]
└─$ su Lok
Password:
$ nano /home/Lok/.bashrc
```

```
alias devread='cat /opt/Bagmaticyber/projects/dev_notes.txt'

┌──(Lok㉿kali)-[~]
└─$ source ~/.bashrc
```

7.3 Audit Trail — Log Every User Login

Use PAM exec to write a line to audit.log whenever any user logs in:

```
┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/pam.d/common-session
```

Add the PAM exec hook to /etc/pam.d/common-session:

```
session optional pam_exec.so /bin/bash -c 'echo "$(date) - $USER logged in" >> /opt/Bagmaticyber/logs/audit.log'
```

7.4 File Integrity Check with md5sum

```
┌──(kali㉿kali)-[~]
└─$ sudo md5sum /opt/Bagmaticyber/README.md | sudo tee /opt/Bagmaticyber/readme_checksum.md5 > /dev/null
```

```
┌──(kali㉿kali)-[~]
└─$ md5sum -c /opt/Bagmaticyber/readme_checksum.md5
/opt/Bagmaticyber/README.md: OK
```

It confirms:

- The checksum stored in readme_checksum.md5
- Matches the **current state** of README.md
- So the file has **NOT been modified**

Quick Reference: Permission Cheat Sheet

Octal	Symbol	Owner	Group	Others	Typical Use
644	rw-r--r--	rw	r	r	Regular files
640	rw-r-----	rw	r	none	Sensitive files (this lab)
444	r--r--r--	r	r	r	Read-only for all (README.md)
770	rwxrwx---	rwx	rwx	none	Team directories (this lab)
2770	rwxrws---	rwx	rwx+s	none	SetGID team directories

Key Commands Summary

Task	Command
Create group	`sudo groupadd <groupname>`
Create user with groups	`sudo useradd -m -g <primary> -G <secondary> <user>`
Change ownership	`sudo chown user:group <file>`
Set permissions + SetGID	`sudo chmod 2770 <dir>`
Add to group	`sudo usermod -aG <group> <user>`
Remove from group	`sudo gpasswd -d <user> <group>`
Edit sudoers safely	`sudo visudo -f /etc/sudoers.d/<name>`
Delete user + home	`sudo userdel -r <user>`
Check group membership	`id <user>  OR  groups <user>`
File integrity check	`sudo md5sum -c <file>.md5`